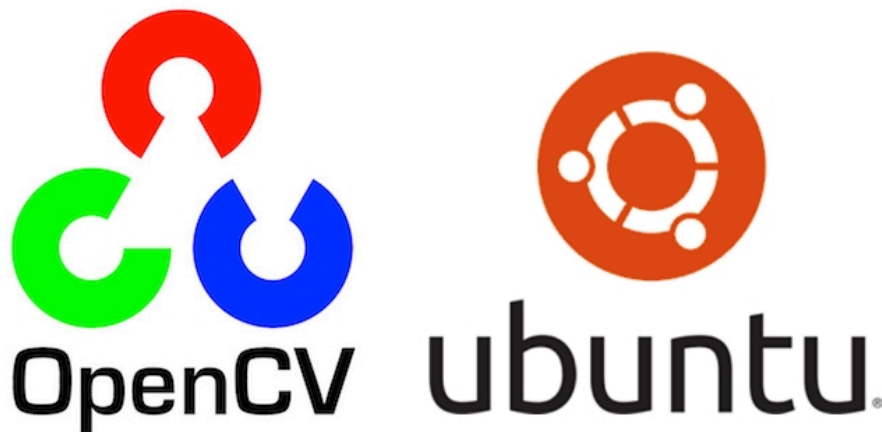


Environment Setup - Installing Open Source CV

If you are using the factory image we provided, manual installation and configuration are not required.

OpenCV (Open Source Computer Vision Library) is an Intel® open-source computer vision library. It consists of a series of C functions and a small number of C++ classes, implementing many general algorithms in image processing and computer vision. OpenCV has cross-platform mid and high-level APIs including more than 300 C functions. It does not depend on other external libraries—although some external libraries can also be used. OpenCV is free for both non-commercial and commercial applications.



As an open-source project for basic computer vision, image processing, and pattern recognition, OpenCV can be directly applied to many fields, making it an ideal tool for secondary development. Especially, the new version of OpenCV provides a large number of Python interfaces, greatly simplifying the calling of OpenCV functions, allowing users to more conveniently use OpenCV to implement various functions.

OpenCV official website: <https://opencv.org/>

It has the following features: (1) Open C/C++ source code (2) Optimized code based on Intel processor instruction set (3) Unified structure and function definition (4) Powerful image and matrix operation capabilities (5) Convenient and flexible user interface (6) Supports both MS-WINDOWS and LINUX platforms

This tutorial uses command installation. Source code installation is also possible but takes longer.

The installation command is as follows:

```
sudo apt-get install python3-opencv
```

When you see the following content, the installation is complete:



4. 192.168.2.152



• MobaXterm Personal Edition v21.1 •
(SSH client, X server and network tools)

- SSH session to `pi@192.168.2.152`
 - Direct SSH : ✓
 - SSH compression : ✓
 - SSH-browser : ✓
 - X11-forwarding : ✓ (remote display is forwarded through SSH)
- For more [info](#), ctrl+click on [help](#) or visit our [website](#).

Linux raspberrypi 6.1.0-rpi7-rpi-2712 #1 SMP PREEMPT Debian 1:6.1.63-1+rpt1 (2023-11-24) aarch64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.

Last login: Tue Dec 19 19:20:56 2023

pi@raspberrypi:~ \$ python3

Python 3.11.2 (main, Mar 13 2023, 12:18:29) [GCC 12.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.

>>> import cv2

>>> █